

In [1]: `import pandas as pd`

In [2]: `emp = pd.read_excel(r"C:\Users\Lenovo\Downloads\Rawdata.xlsx")`

In [3]: `emp`

Out[3]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [4]: `emp.columns`

Out[4]: `Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')`

In [5]: `emp.shape`

Out[5]: `(6, 6)`

In [6]: `emp.head()`

Out[6]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year

In [7]: `emp.tail()`

Out[7]:

	Name	Domain	Age	Location	Salary	Exp
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [8]: `emp.isnull()`

Out[8]:

	Name	Domain	Age	Location	Salary	Exp
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	True	True	False	False
3	False	False	True	False	False	True
4	False	False	False	True	False	False
5	False	False	False	False	False	False

In [9]: `emp.isna()`

Out[9]:

	Name	Domain	Age	Location	Salary	Exp
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	True	True	False	False
3	False	False	True	False	False	True
4	False	False	False	True	False	False
5	False	False	False	False	False	False

In [10]: `emp.isna().sum()`

Out[10]:

```
Name      0
Domain    0
Age        2
Location   2
Salary     0
Exp        1
dtype: int64
```

In [11]: `emp.isna().sum().sum()`

Out[11]: 5

In [12]: `emp.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        6 non-null      object
1   Domain      6 non-null      object
2   Age         4 non-null      object
3   Location    4 non-null      object
4   Salary      6 non-null      object
5   Exp         5 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes
```

In [13]: emp['Domain']

Out[13]: 0 Datascience#\$
1 Testing
2 Dataanalyst^^#
3 Ana^^lytics
4 Statistics
5 NLP
Name: Domain, dtype: object

In [14]: emp

Out[14]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [18]: emp['Name'] = emp['Name'].str.replace(r'\W', '', regex=True)

In [19]: emp['Name']

Out[19]: 0 Mike
1 Teddy
2 Umar
3 Jane
4 Uttam
5 Kim
Name: Name, dtype: object

In [22]: emp['Domain'] = emp['Domain'].str.replace(r'\W', '', regex=True)

In [23]: emp['Domain']

Out[23]: 0 Datascience
1 Testing
2 Dataanalyst
3 Analytics
4 Statistics
5 NLP
Name: Domain, dtype: object

In [24]: emp['Age']

Out[24]: 0 34 years
1 45' yr
2 NaN
3 NaN
4 67-yr
5 55yr
Name: Age, dtype: object

```
In [46]: emp['Age'] = emp['Age'].str.extract('(\d+)')
emp['Age']
```

```
Out[46]: 0      34
         1      45
         2      NaN
         3      NaN
         4      67
         5      55
         Name: Age, dtype: object
```

```
In [47]: emp['Location']
```

```
Out[47]: 0      Mumbai
         1    Bangalore
         2         NaN
         3    Hyderabad
         4         NaN
         5        Delhi
         Name: Location, dtype: object
```

```
In [48]: emp['Location'] = emp['Location'].str.replace(r'\W', '', regex=True)
emp['Location']
```

```
Out[48]: 0      Mumbai
         1    Bangalore
         2         NaN
         3    Hyderabad
         4         NaN
         5        Delhi
         Name: Location, dtype: object
```

```
In [49]: emp['Salary']
```

```
Out[49]: 0      5000
         1     10000
         2     15000
         3     20000
         4     30000
         5     60000
         Name: Salary, dtype: object
```

```
In [50]: emp['Salary'] = emp['Salary'].str.replace(r'\W', '', regex=True)
emp['Salary']
```

```
Out[50]: 0      5000
         1     10000
         2     15000
         3     20000
         4     30000
         5     60000
         Name: Salary, dtype: object
```

```
In [30]: emp['Exp']
```

```
Out[30]: 0      2+
         1      <3
         2      4> yrs
         3      NaN
         4      5+ year
         5      10+
         Name: Exp, dtype: object
```

```
In [51]: emp['Exp'] = emp['Exp'].str.extract('(\d+)')
         emp['Exp']
```

```
<>:1: SyntaxWarning: invalid escape sequence '\d'
<>:1: SyntaxWarning: invalid escape sequence '\d'
C:\Users\Lenovo\AppData\Local\Temp\ipykernel_4728\4181881072.py:1: SyntaxWarning:
invalid escape sequence '\d'
emp['Exp'] = emp['Exp'].str.extract('(\d+)')
```

```
Out[51]: 0      2
         1      3
         2      4
         3      NaN
         4      5
         5      10
         Name: Exp, dtype: object
```

```
In [52]: emp
```

```
Out[52]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [63]: clean_data = emp.copy()
```

```
In [64]: clean_data
```

```
Out[64]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [65]: clean_data['Age']
```

```
Out[65]: 0      34
         1      45
         2      NaN
         3      NaN
         4      67
         5      55
         Name: Age, dtype: object
```

```
In [66]: clean_data.isna().sum()
```

```
Out[66]: Name      0
         Domain    0
         Age       2
         Location   2
         Salary    0
         Exp       1
         dtype: int64
```

```
In [67]: clean_data['Age']
```

```
Out[67]: 0      34
         1      45
         2      NaN
         3      NaN
         4      67
         5      55
         Name: Age, dtype: object
```

```
In [68]: import numpy as np
```

```
In [70]: clean_data['Age'] = clean_data['Age'].fillna(np.mean(pd.to_numeric(clean_data['Age'],
clean_data['Age']
```

```
Out[70]: 0      34
         1      45
         2     50.25
         3     50.25
         4      67
         5      55
         Name: Age, dtype: object
```

```
In [71]: clean_data['Exp'] = clean_data['Exp'].fillna(np.mean(pd.to_numeric(clean_data['Exp'],
clean_data['Exp']
```

```
In [72]: clean_data['Exp']
```

```
Out[72]: 0      2
         1      3
         2      4
         3     4.8
         4      5
         5     10
         Name: Exp, dtype: object
```

```
In [75]: clean_data['Location'] = clean_data['Location'].fillna(clean_data['Location'].mode().iloc[0])
```

```
In [77]: clean_data['Location']
```

```
Out[77]: 0      Mumbai
        1      Bangalore
        2      Bangalore
        3      Hyderabad
        4      Bangalore
        5      Delhi
        Name: Location, dtype: object
```

```
In [78]: clean_data
```

```
Out[78]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50.25	Bangalore	15000	4
3	Jane	Analytics	50.25	Hyderabad	20000	4.8
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [79]: clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        6 non-null      object
1   Domain      6 non-null      object
2   Age         6 non-null      object
3   Location    6 non-null      object
4   Salary      6 non-null      object
5   Exp         6 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes
```

```
In [81]: clean_data['Age'] = clean_data['Age'].astype(int)
        clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        6 non-null      object
1   Domain      6 non-null      object
2   Age         6 non-null      int32
3   Location    6 non-null      object
4   Salary      6 non-null      object
5   Exp         6 non-null      object
dtypes: int32(1), object(5)
memory usage: 396.0+ bytes
```

```
In [82]: clean_data['Exp'] = clean_data['Exp'].astype(int)
        clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        6 non-null      object
1   Domain      6 non-null      object
2   Age         6 non-null      int32
3   Location    6 non-null      object
4   Salary      6 non-null      object
5   Exp         6 non-null      int32
dtypes: int32(2), object(4)
memory usage: 372.0+ bytes
```

```
In [87]: clean_data['Name'] = clean_data['Name'].astype('category')
clean_data['Domain'] = clean_data['Domain'].astype('category')
clean_data['Location'] = clean_data['Location'].astype('category')
clean_data['Salary'] = clean_data['Salary'].astype('category')

clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        6 non-null      category
1   Domain      6 non-null      category
2   Age         6 non-null      int32
3   Location    6 non-null      category
4   Salary      6 non-null      category
5   Exp         6 non-null      int32
dtypes: category(4), int32(2)
memory usage: 1.0 KB
```

```
In [88]: clean_data
```

```
Out[88]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderbad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [89]: clean_data.to_csv('clean_data.csv')
```

```
In [91]: import os
os.getcwd()
```

```
Out[91]: 'C:\\Users\\Lenovo'
```

```
In [92]: clean_data
```


Out[92]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderbad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [93]: `import matplotlib.pyplot as plt`
`import seaborn as sns`

In [94]: `import warnings`
`warnings.filterwarnings('ignore')`

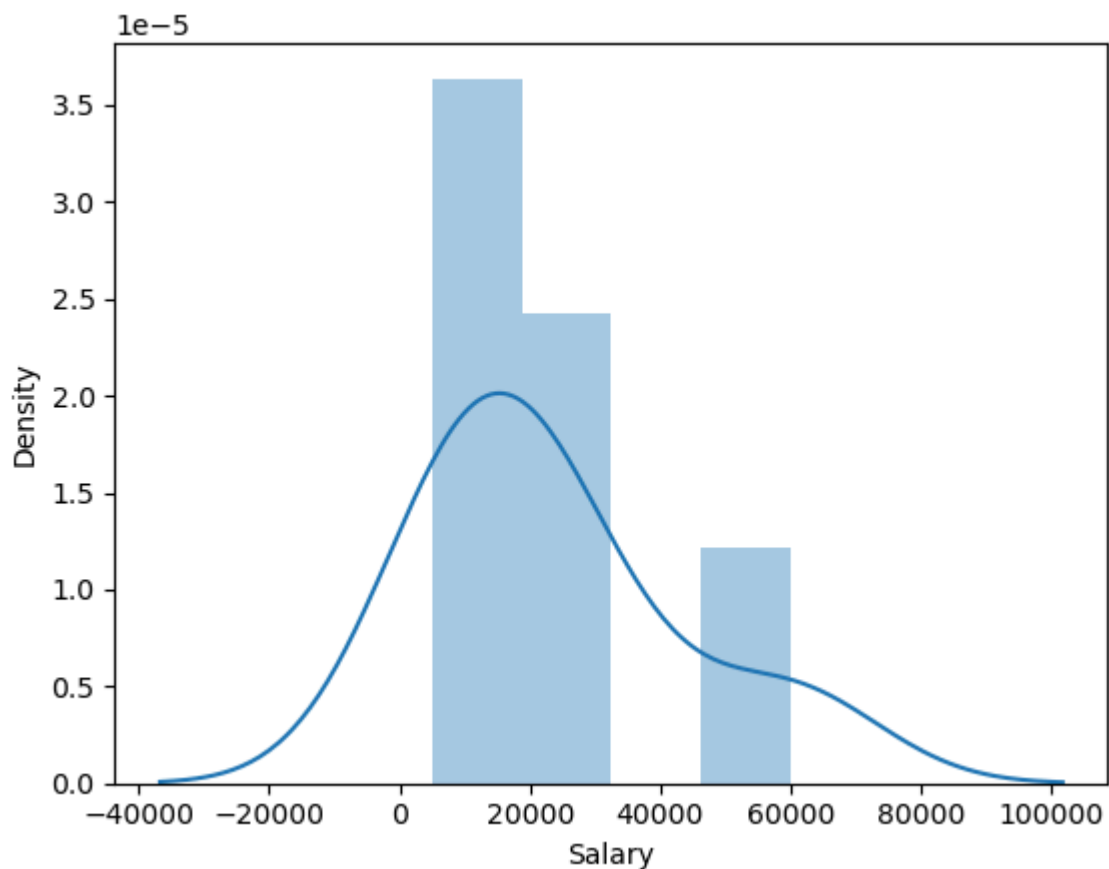
In [95]: `clean_data['Salary']`

Out[95]:

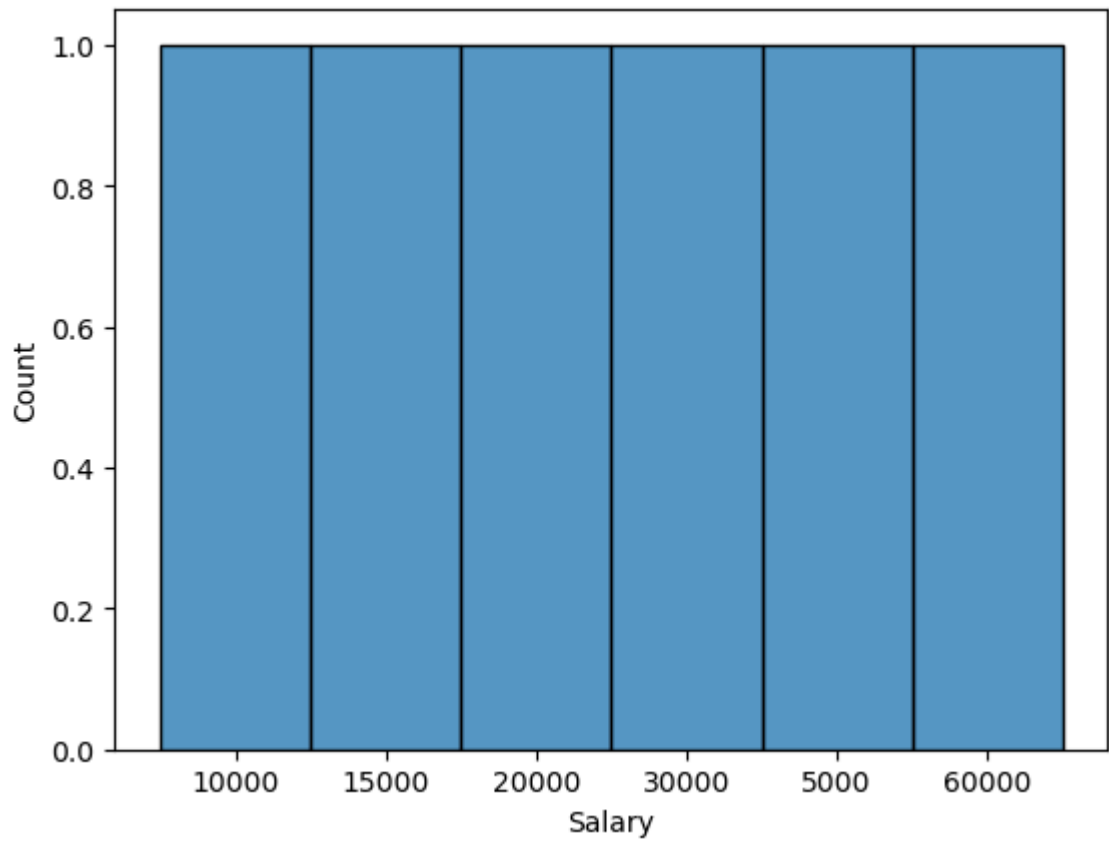
0	5000
1	10000
2	15000
3	20000
4	30000
5	60000

Name: Salary, dtype: category
Categories (6, object): ['10000', '15000', '20000', '30000', '5000', '60000']

In [101... `vis1 = sns.distplot(clean_data['Salary'])`



```
In [106... vis2 = sns.histplot(clean_data['Salary'])
```



```
In [112... vis3 = sns.lmplot(data=clean_data, x='Exp', y='Salary')
```

```

-----
TypeError                                Traceback (most recent call last)
Cell In[112], line 1
----> 1 vis3 = sns.lmplot(data=clean_data, x='Exp', y='Salary')

File ~\anaconda3\Anaconda\Lib\site-packages\seaborn\regression.py:651, in lmplot
(data, x, y, hue, col, row, palette, col_wrap, height, aspect, markers, sharex, s
harey, hue_order, col_order, row_order, legend, legend_out, x_estimator, x_bins,
x_ci, scatter, fit_reg, ci, n_boot, units, seed, order, logistic, lowess, robust,
logx, x_partial, y_partial, truncate, x_jitter, y_jitter, scatter_kws, line_kws,
facet_kws)
    642 # Draw the regression plot on each facet
    643 regplot_kws = dict(
    644     x_estimator=x_estimator, x_bins=x_bins, x_ci=x_ci,
    645     scatter=scatter, fit_reg=fit_reg, ci=ci, n_boot=n_boot, units=units,
    (...)
    649     scatter_kws=scatter_kws, line_kws=line_kws,
    650 )
--> 651 facets.map_dataframe(regplot, x=x, y=y, **regplot_kws)
    652 facets.set_axis_labels(x, y)
    654 # Add a legend

File ~\anaconda3\Anaconda\Lib\site-packages\seaborn\axisgrid.py:825, in FacetGrid.map_dataframe(self, func, *args, **kwargs)
    822     kwargs["data"] = data_ijk
    824     # Draw the plot
--> 825     self._facet_plot(func, ax, args, kwargs)
    827 # For axis labels, prefer to use positional args for backcompat
    828 # but also extract the x/y kwargs and use if no corresponding arg
    829 axis_labels = [kwargs.get("x", None), kwargs.get("y", None)]

File ~\anaconda3\Anaconda\Lib\site-packages\seaborn\axisgrid.py:854, in FacetGrid._facet_plot(self, func, ax, plot_args, plot_kws)
    852     plot_args = []
    853     plot_kws["ax"] = ax
--> 854 func(*plot_args, **plot_kws)
    856 # Sort out the supporting information
    857 self._update_legend_data(ax)

File ~\anaconda3\Anaconda\Lib\site-packages\seaborn\regression.py:775, in regplot
(data, x, y, x_estimator, x_bins, x_ci, scatter, fit_reg, ci, n_boot, units, seed,
order, logistic, lowess, robust, logx, x_partial, y_partial, truncate, dropna,
x_jitter, y_jitter, label, color, marker, scatter_kws, line_kws, ax)
    773 scatter_kws["marker"] = marker
    774 line_kws = {} if line_kws is None else copy.copy(line_kws)
--> 775 plotter.plot(ax, scatter_kws, line_kws)
    776 return ax

File ~\anaconda3\Anaconda\Lib\site-packages\seaborn\regression.py:384, in _RegressionPlotter.plot(self, ax, scatter_kws, line_kws)
    381     self.scatterplot(ax, scatter_kws)
    383 if self.fit_reg:
--> 384     self.lineplot(ax, line_kws)
    386 # Label the axes
    387 if hasattr(self.x, "name"):

File ~\anaconda3\Anaconda\Lib\site-packages\seaborn\regression.py:429, in _RegressionPlotter.lineplot(self, ax, kws)
    427 """Draw the model."""
    428 # Fit the regression model

```

```
--> 429 grid, yhat, err_bands = self.fit_regression(ax)
      430 edges = grid[0], grid[-1]
      432 # Get set default aesthetics
```

File ~\anaconda3\Anaconda\Lib\site-packages\seaborn\regression.py:229, in _RegressionPlotter.fit_regression(self, ax, x_range, grid)

```
      227 yhat, yhat_boots = self.fit_logx(grid)
      228 else:
--> 229 yhat, yhat_boots = self.fit_fast(grid)
      231 # Compute the confidence interval at each grid point
      232 if ci is None:
```

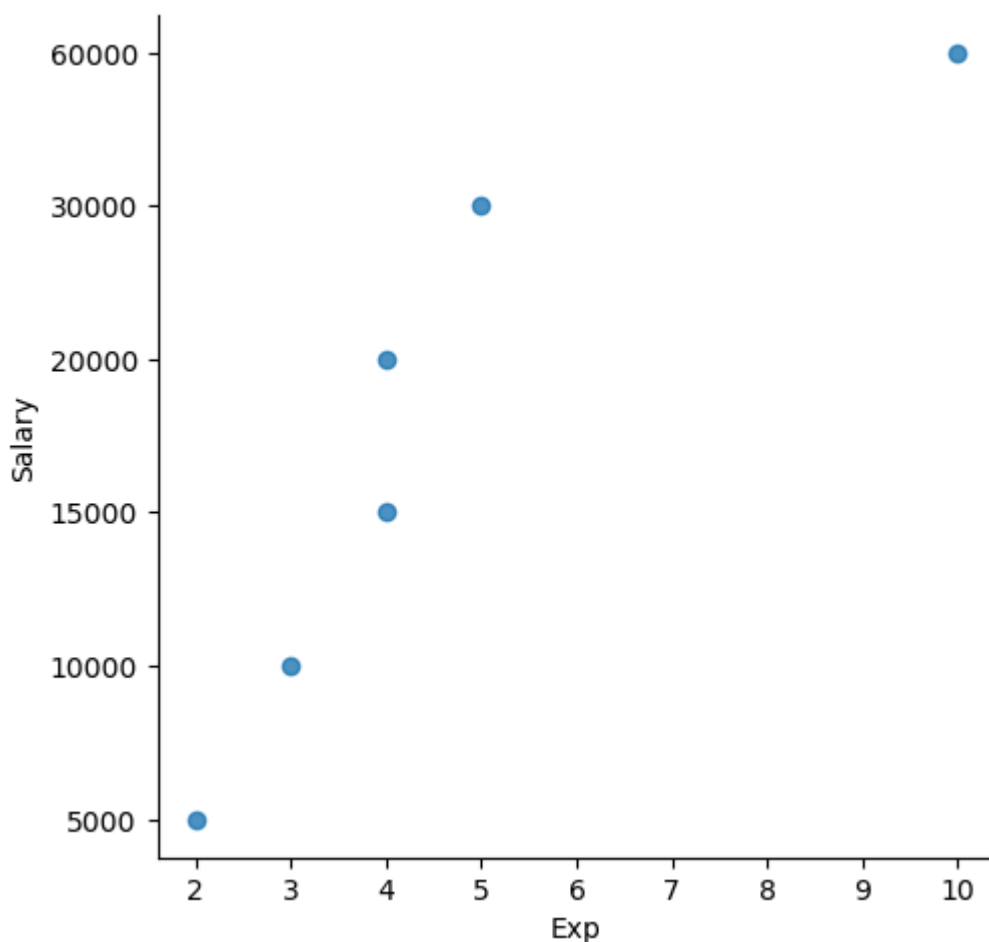
File ~\anaconda3\Anaconda\Lib\site-packages\seaborn\regression.py:246, in _RegressionPlotter.fit_fast(self, grid)

```
      244 X, y = np.c_[np.ones(len(self.x)), self.x], self.y
      245 grid = np.c_[np.ones(len(grid)), grid]
--> 246 yhat = grid.dot(reg_func(X, y))
      247 if self.ci is None:
      248     return yhat, None
```

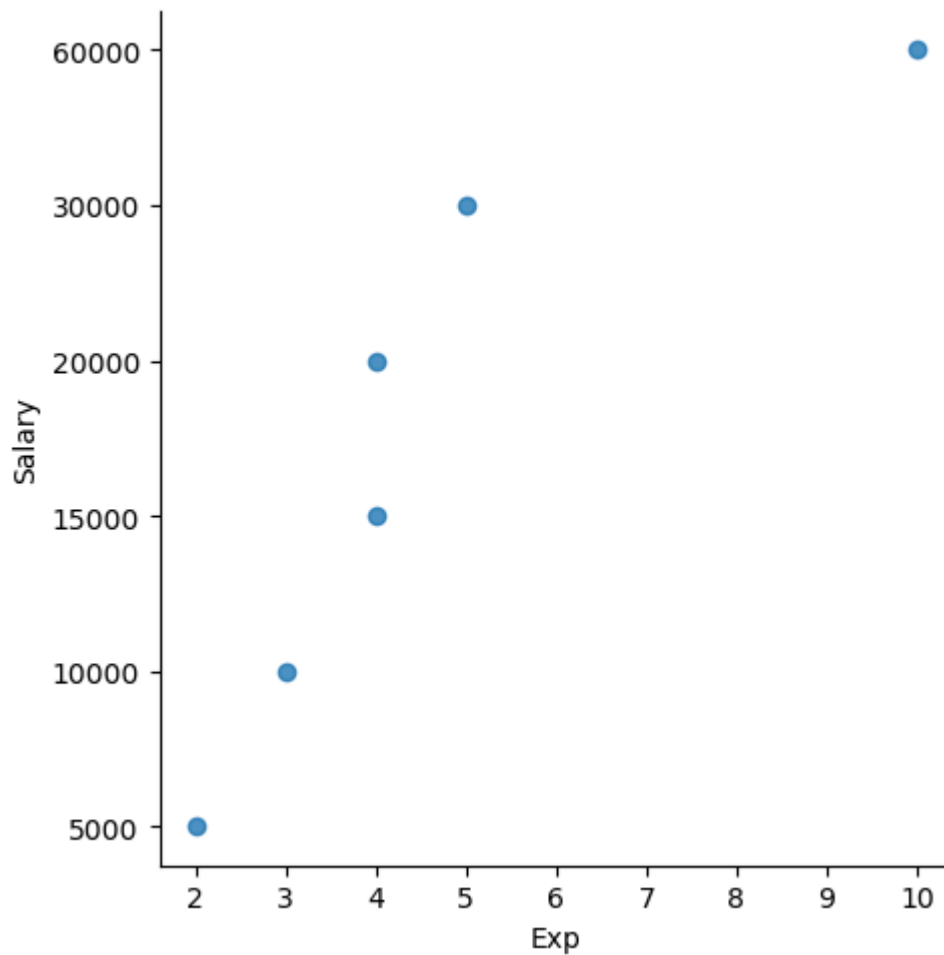
File ~\anaconda3\Anaconda\Lib\site-packages\seaborn\regression.py:242, in _RegressionPlotter.fit_fast.<locals>.reg_func(_x, _y)

```
      241 def reg_func(_x, _y):
--> 242     return np.linalg.pinv(_x).dot(_y)
```

TypeError: can't multiply sequence by non-int of type 'float'



```
In [114... vis4 = sns.lmplot(data=clean_data, x='Exp', y='Salary', fit_reg=False)
```



```
In [115... clean_data[:]
```

```
Out[115... 
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderabad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [116... clean_data[0:6:2]
```

```
Out[116... 
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
2	Umar	Dataanalyst	50	Bangalore	15000	4
4	Uttam	Statistics	67	Bangalore	30000	5

```
In [117... clean_data[:, :-1]
```

Out[117...

	Name	Domain	Age	Location	Salary	Exp
5	Kim	NLP	55	Delhi	60000	10
4	Uttam	Statistics	67	Bangalore	30000	5
3	Jane	Analytics	50	Hyderabad	20000	4
2	Umar	Dataanalyst	50	Bangalore	15000	4
1	Teddy	Testing	45	Bangalore	10000	3
0	Mike	Datascience	34	Mumbai	5000	2

In [118...

clean_data.columns

Out[118...

Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')

In [119...

X_iv = clean_data[['Name', 'Domain', 'Age', 'Location', 'Exp']]

In [121...

X_iv

Out[121...

	Name	Domain	Age	Location	Exp
0	Mike	Datascience	34	Mumbai	2
1	Teddy	Testing	45	Bangalore	3
2	Umar	Dataanalyst	50	Bangalore	4
3	Jane	Analytics	50	Hyderabad	4
4	Uttam	Statistics	67	Bangalore	5
5	Kim	NLP	55	Delhi	10

In [122...

y_dv = clean_data[['Salary']]

In [123...

y_dv

Out[123...

	Salary
0	5000
1	10000
2	15000
3	20000
4	30000
5	60000

In [124...

emp

Out[124...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderabad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [125...

clean_data

Out[125...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderabad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [126...

X_iv

Out[126...

	Name	Domain	Age	Location	Exp
0	Mike	Datascience	34	Mumbai	2
1	Teddy	Testing	45	Bangalore	3
2	Umar	Dataanalyst	50	Bangalore	4
3	Jane	Analytics	50	Hyderabad	4
4	Uttam	Statistics	67	Bangalore	5
5	Kim	NLP	55	Delhi	10

In [127...

y_dv

Out[127...

Salary**0** 5000**1** 10000**2** 15000**3** 20000**4** 30000**5** 60000

In [128...

clean_data

Out[128...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderbad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [129...

imputatios = pd.get_dummies(clean_data)

In [130...

imputatios

Out[130...

	Age	Exp	Name_Jane	Name_Kim	Name_Mike	Name_Teddy	Name_Umar	Name_U
0	34	2	False	False	True	False	False	
1	45	3	False	False	False	True	False	
2	50	4	False	False	False	False	True	
3	50	4	True	False	False	False	False	
4	67	5	False	False	False	False	False	
5	55	10	False	True	False	False	False	

6 rows × 24 columns



In [131...

clean_data

Out[131...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderbad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [134...

imputatios

Out[134...

	Age	Exp	Name_Jane	Name_Kim	Name_Mike	Name_Teddy	Name_Umar	Name_U
0	34	2	False	False	True	False	False	
1	45	3	False	False	False	True	False	
2	50	4	False	False	False	False	True	
3	50	4	True	False	False	False	False	
4	67	5	False	False	False	False	False	
5	55	10	False	True	False	False	False	

6 rows × 24 columns



In []: